



Handout Garmin G1000 Part 3 Operation

**Part 3a: VFR
Part 3b: IFR**

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Garmin G1000



G1000 Operation (VFR)



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Data Field Settings



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MFD data fields Aux Page Group



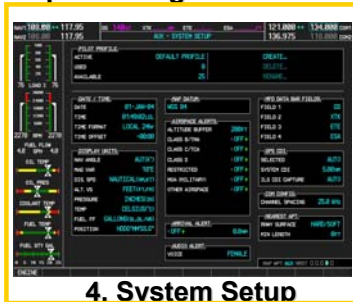
1. Trip Planning



2. Utility



3. GPS Status



4. System Setup



5. System Status



MFD NAV status box



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Autozoom PFD and MFD



Auto Zoom:
Off, MFD Only, PFD Only, All On

Manual Range Override
Expiration Time



Maximum Look Forward Time

Minimum Look Forward Time



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Data Input



- via Input Panel
- turning the small FMS knob brings up the input panel !



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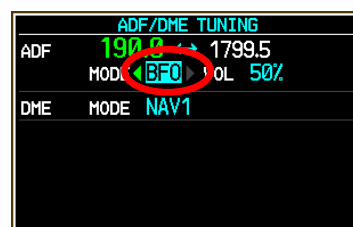
Data Input

- Selecting a Choice
- turning the small FMS knob brings up a selection window !



Data Input

- When a green triangle appears near a field:
 - Field value may be changed by turning the small FMS knob in direction of the triangle!



Flight Plan Composition



Flight Plan Composition

- 2 places to build a flight plan:
 - Active Flight Plan (page 1)
 - FPL not saved automatically
 - Flight Plan Catalog (page 2)
 - FPL saved automatically



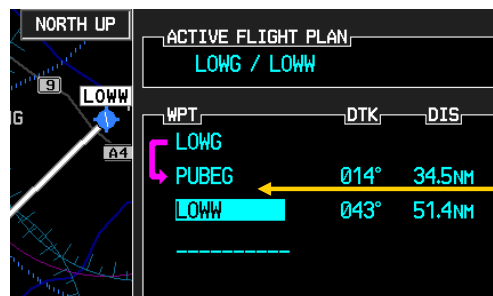
Flight Plan Composition

- 2 methods to build a flight plan:
 - „sequentially“
 - DEP AD
 - waypoints
 - DEST AD
 - „fill in method“
 - DEP AD
 - DEST AD
 - waypoints inserted later



Flight Plan Composition

- Inserting a waypoint
 - A new waypoint is inserted above the highlighted waypoint



SNU
inserted here



Flight Plan Composition

- Remember:
- „In the Flight Plan everything happens ABOVE the cursor!“



Flight plan import/export

17.80 TOP GS 188KT DTK 147° TRK 147° ETE 02:01 133.300 ↔ 121.900 COM1
15.50 STJ FPL - FLIGHT PLAN CATALOG 128.200 118.200 COM2

NORTH UP

FLIGHT PLAN CATALOG	
USED	4
EMPTY	95

FLIGHT PLAN LIST	
1	KMHC / KCOS
2	KMHC / KCOS
3	KMHC / KCOS
4	KIXD / KMCI
5	_____ / _____
6	_____ / _____
7	_____ / _____
8	_____ / _____
9	_____ / _____
10	_____ / _____
11	_____ / _____
12	_____ / _____

FLIGHT PLAN INFO:

DEPARTURE KMHC
DESTINATION KCOS
TOTAL DISTANCE 618nm
ENROUTE SAFE ALT _____FT

MAP NEW ACTIVE INVERT EDIT COPY DELETE IMPORT EXPORT



Flight plan import/export

FLIGHT PLAN CATALOG
USED 4 EMPTY 95

FLIGHT PLAN LIST

1	KMCC / KCOS
2	KMCC / KCOS
3	KMCC / KCOS
4	KIXD / KNCI
5	----- / -----
6	----- / -----
7	----- / -----
8	----- / -----
9	----- / -----
10	----- / -----
11	----- / -----
12	----- / -----

FLIGHT PLAN INFO

DEPARTURE KIXD
DESTINATION KNCI
TOTAL DISTANCE 677NM
ENROUTE SAFE ALT 4200FT

FPL

COPY DELETE IMPORT EXPORT

FLIGHT PLAN IMPORTING

FILE LIST
Select file to load into stored flight plan 4:

0000.fpl
0001.fpl
0002.fpl
0003.fpl
0004.fpl
0005.fpl
0006.fpl
0007.fpl
0008.fpl
0009.fpl
0010.fpl
0011.fpl
0012.fpl

FILE DETAILS

NAME KIXD / TBE
CREATE ON 14-NOV-00 14:52:30UTC
CREATE BY Unknown

IMPORT RESULTS

FLIGHT PLAN IMPORTING

FILE LIST
Select file to load into stored flight plan 4:

0000.fpl
0001.fpl
0002.fpl
0003.fpl
0004.fpl
0005.fpl
0006.fpl
0007.fpl
0008.fpl
0009.fpl
0010.fpl
0011.fpl
0012.fpl

FILE DETAILS

NAME KIXD / TBE
CREATE ON 14-NOV-00 14:52:30UTC
CREATE BY Unknown

IMPORT RESULTS
Flight plan successfully imported.

OK

FLIGHT PLAN EXPORTING

FILE NAME
Export stored flight plan 4 to file name:

0005.fpl

EXPORT RESULTS

OK

10 ----- / -----
11 ----- / -----
12 ----- / -----

FLIGHT PLAN INFO

DEPARTURE KIXD
DESTINATION KNCI
TOTAL DISTANCE 677NM
ENROUTE SAFE ALT 4200FT

FLIGHT PLAN EXPORTING

FILE NAME
Export stored flight plan 4 to file name:

0005.fpl

EXPORT RESULTS
Flight plan successfully exported.

OK

10 ----- / -----
11 ----- / -----
12 ----- / -----

FLIGHT PLAN INFO

DEPARTURE KIXD
DESTINATION KNCI
TOTAL DISTANCE 677NM
ENROUTE SAFE ALT 4200FT



Intentionally blank



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Waypoint Entry Logic

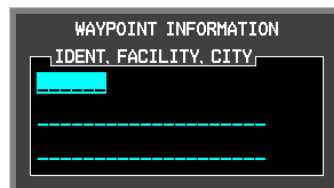


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WPT entry logic



Small FMS knob clockwise:
Enter WPT
Small FMS knob counter-clockwise:
WPT Submenu:



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Practical Hints

■ Softkeys

- Pay attention to the softkeys !
- If there is no softkey for your desired action:
- Use the menu



Softkeys #1

To activate a leg

You want to activate
LOWW - LNZ

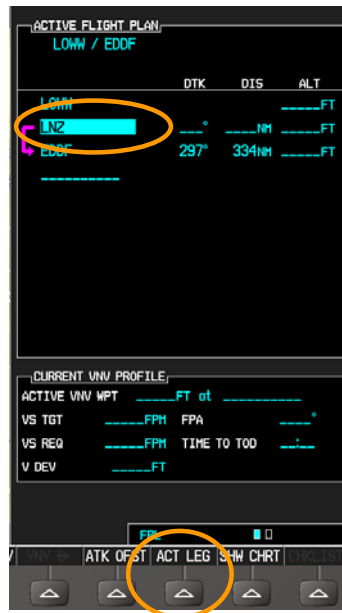
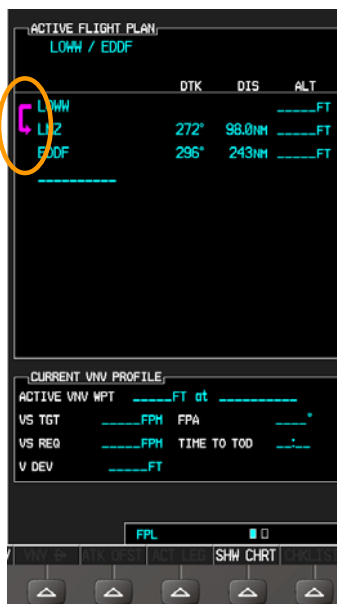


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Softkeys #1



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Softkeys #2



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Softkeys #3

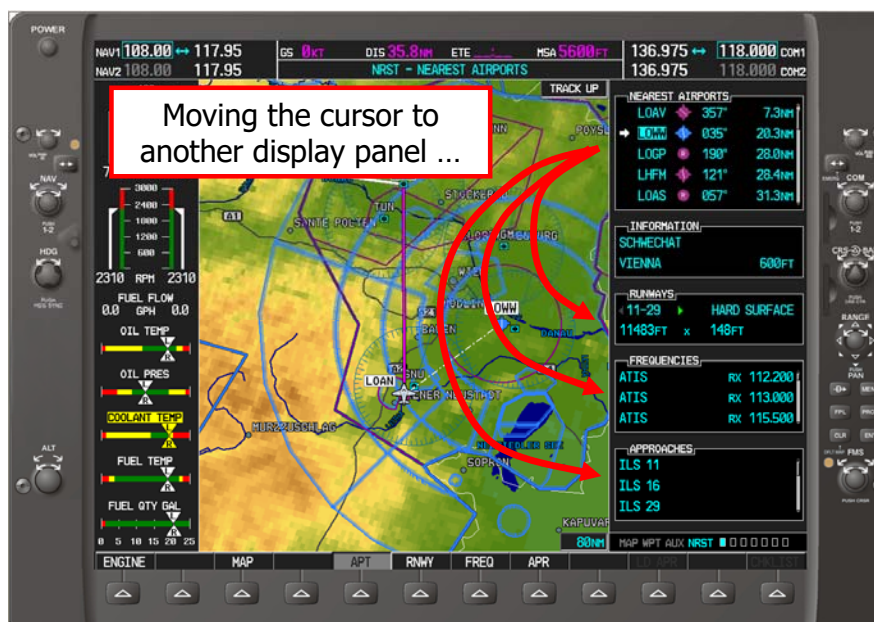


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Softkeys #3

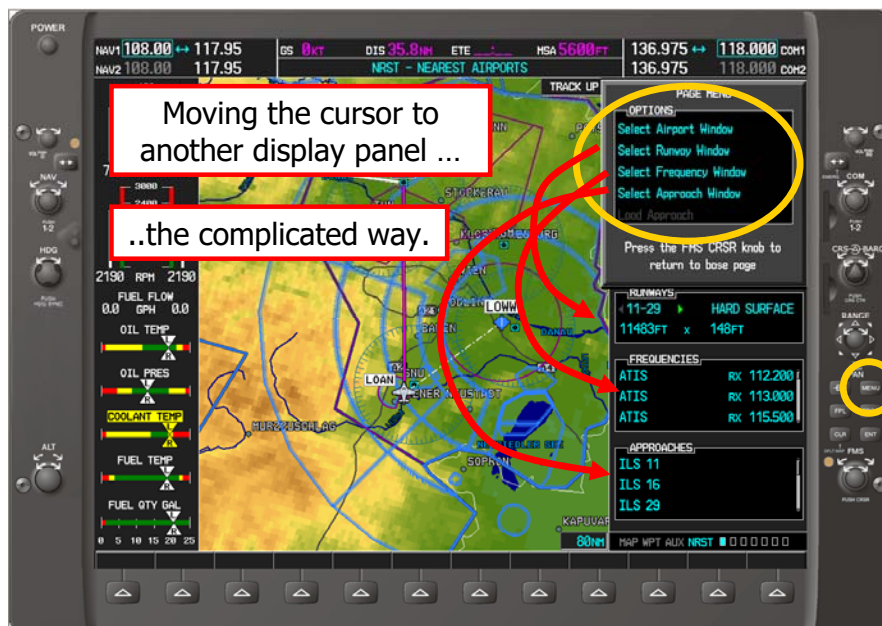


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Softkeys #3



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G1000 Operation (IFR)



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IFR with the G1000



- For flying IFR with the G1000 you need:
 - a solid foundation of instrument flying
 - to be organized and professional
 - to follow procedures and checklists
 - to feel comfortable with instrument approaches, last-minute reroutings, holdings and system failures



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IFR with the G1000

- Especially for single pilot operation
 - autopilot and FMS are not just convenient and helpful ways to ease pilot workload
 - they are critical and required for a safe operation



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G1000 Limitations

- IFR only with valid database, or each WPT has to be verified
- Instrument approaches only in approach mode and RAIM must be available at FAF
- VOR/ILS approaches: VOR/ILS data have to be on the CDI display
- RNAV (GPS) approaches must utilize GPS sensor
- GPS guidance approved only for „RNAV“, „GPS“ or „GPS overlay“-approaches



G1000 Limitations



- Alternate APT: APCH other than GPS must be available
- VNAV: advisory only; Pilot's altimeter is primary reference
- Compulsory settings
 - DIS, SPD nm kt
 - ALT, VS ft fpm
 - POSITION deg – min



G1000 Limitations



The availability of SafeTaxi®, ChartView, or FliteCharts® in electronic form on the G1000 is for information purposes only, it is still mandatory to carry another source of charts on-board the airplane.



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Jeppesen Data Base Limitations



Jeppesen Limitations



- „Uncodeable“ procedures are not included in the Jeppesen Master Database.
- Stepdown fixes between the FAF and MAP are not included in navigation database.
- Regardless of what is shown in the electronic chart, all legs of the procedure on the paper chart have to be flown as charted.



Jeppesen Limitations

- You may not be authorized to fly all procedures in your database. If you don't have a paper chart for it, you are not authorized to fly it.
- Some categories of controlled airspace are not in your database – Class A, E and F is not included.



Jeppesen Limitations

- Not all altitudes are in the database.
- Not included are:
 - MDA
 - DA/DH
 - MOCA
 - MRA
 - MSA
 - MCA
 - MAA



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GPS CDI Sub-Modes



The mode is indicated here



GPS CDI Sub-Modes



Possible modes are:

AUTO SEQUENCING

OBS

Omni Bearing Select

SUSP

Suspend



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GPS CDI Sub-Modes



AUTO SEQUENCING

blank

not active (black)



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GPS CDI Sub-Modes



Omni Bearing Select

OBS

active (grey)



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GPS CDI Sub-Modes

- Use of the OBS mode:
 - designated course to a WPT
 - Holding Pattern
 - RWY centerline



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GPS CDI Sub-Modes



SUSPEND

SUSP

active (grey)

to deactivate SUSPEND

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GPS CDI Sub-Modes

- SUSP mode is activated:
 - when passing the MAPt
 - in the Holding Pattern



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Approach Types

HSI Annunciation	Description	Example on HSI
LNAV	GPS approach using published LNAV minima	Approach Type - LNAV - LNAV+V - L/VNAV - LPV
LNAV+V	GPS approach using published LNAV minima. Advisory vertical guidance is provided	
L/VNAV	GPS approach using published LNAV/VNAV minima	
LPV	GPS approach using published LPV minima	

LNAV/VNAV



Flight phase indication

Flight Phase	Annunciation*	Automatic CDI Full-scale Deflection
Departure	DPRT	0.3 nm
Terminal	TERM	1.0 nm
Enroute	ENR	2.0 nm
Oceanic	OCN	2.0 nm
Approach (Non-precision)	LNAV	1.0 nm decreasing to 350 feet
Approach (Non-precision with Vertical Guidance)	LNAV + V	SBAS necessary
Approach (LNAV/VNAV)	L/VNAV	1.0 nm decreasing to a specified course width
Approach (LPV)	LPV	
Missed Approach	MAPR	0.3 nm



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Some Limitations

AFM Supplement A33

- NAV information must be WGS84
- Database must be current
- Manual entry of waypoints (LAT/LONG, PBD) prohibited
- When SBAS coverage confirmed: no WFDE or RAIM prediction necessary
- SBAS must be enabled



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GPS status page (AUX 3)



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Flight Plan Composition



Flight Plan Composition

- 2 places to build a flight plan:
 - Active Flight Plan (page 1)
 - FPL not saved automatically
 - Flight Plan Catalog (page 2)
 - FPL saved automatically



Flight Plan Composition

- 2 methods to build a flight plan:
 - „sequentially“
 - DEP AD
 - SID
 - waypoints, airways
 - (STAR) (4-letter DEST inserted automatically)
 - APCH (4-letter DEST disappears)
 - „fill in method“
 - DEP AD
 - DEST AD
 - SID, waypoints, airways, STAR, APCH inserted later
 - **STAR and APCH treated differently!**



Flight Plan Composition

- Approach vs Arrival (STAR)
 - „Approach“ and „STAR“ are treated differently !
 - Approach:
 - adds approach procedure to the flight plan (at the end) in „standby“, ready for quick activation
 - Arrival (STAR):
 - immediately part of the active flight plan
 - might therefore create the „zig-zag“ problem !



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Standard Instrument Departures



Check the procedure !



DEPARTURE ROUTE DESCRIPTION

TAKEOFF RWYS 1L/R: Climb heading 003°, or as assigned by ATC, Thence...

TAKEOFF RWY 10: Climb heading 093°, Thence...

TAKEOFF RWY 19L: **JETS:** Climb heading 213°, Thence... **ALL OTHERS:**

Climb heading 183° to cross PIE R-116 at or above 2600, Thence...

TAKEOFF RWY 19R: **JETS:** Climb heading 203°, Thence... **ALL OTHERS:**

Climb heading 183° to cross PIE R-116 at or above 2600, Thence...

TAKEOFF RWY 28: Climb heading 273°, Thence...

Expect radar vectors to join filed/assigned route. **JETS:** Maintain 6000. **ALL OTHERS:** Maintain 3000. Expect clearance to filed altitude/flight level within 10 minutes after departure.



Check the procedure !

- Always crosscheck the displayed route information with the published procedure
- Do not load a SID or STAR that contains JETS-specific instructions (or other procedure that are not applicable to your operation or ATC clearance)
- In cases where more than one aircraft type-specific instruction is provided, only the "JETS" aircraft type-specific instructions are encoded in the Navigation Database



Watch your first leg !

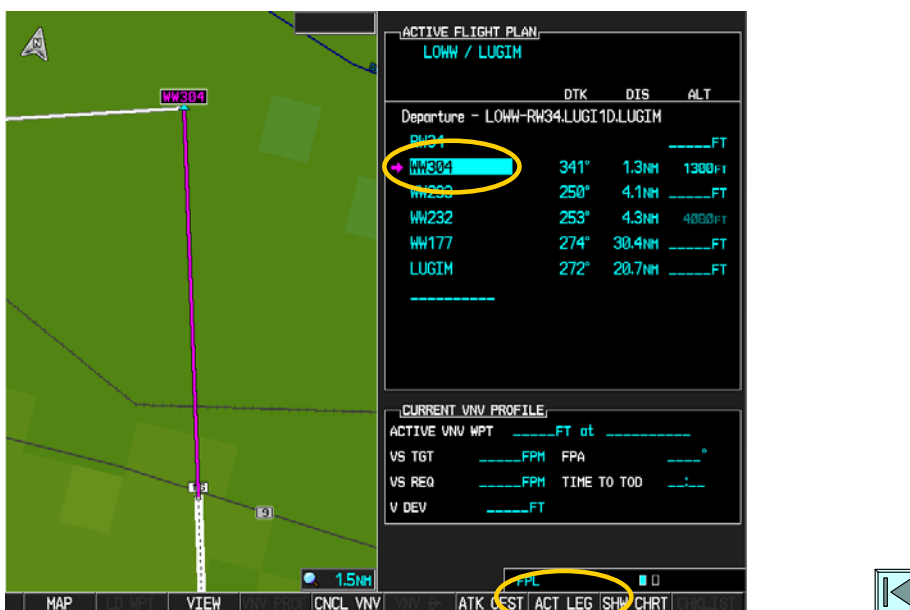
- Depending on your position, your active leg might „jump“ down in the flight plan
- You will have to activate your first leg!



Watch your first leg !



Watch your first leg !



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Airway display

- The software in the DA42 does not show all waypoints along an airway (some waypoints are not shown, they are „Optional“ waypoints)
- Waypoints are shown according the following criteria:
 - VOR and NDB Airway:
 - VORs and NDBs will always be included
 - VOR airways:
 - a waypoint will be included if that waypoint is necessary for a course change
 - RNAV routes
 - Only “Essential” waypoints will be included. “Essential” waypoints are waypoints needed for a course change or defining an intersection with another airway
 - All Air Traffic Control compulsory reporting points (even if no course change or not “Essential”)



Airway display

ACTIVE FLIGHT PLAN LOWW / EDDF			
	DTK	DIS	ALT
SITNI	269°	31.1NM	_____FT
Airway - UL851.LNZ			
LNZ	287°	31.2NM	_____FT
Airway - T161.PSA			
SUBEN	290°	33.1NM	_____FT
ABABI	290°	16.7NM	_____FT
AKINI	289°	35.1NM	_____FT
NIMDI	278°	19.7NM	_____FT
ERNAS	277°	16.6NM	_____FT
REDNI	316°	19.2NM	_____FT
ROTAX	314°	20.7NM	_____FT

Airway designator



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Airway display

ACTIVE FLIGHT PLAN LOWW / EDDF	
ABABI	
AKINI	
NIMDI	
ERNAS	
REDNI	
ROTAX	
TALAK	
LEPSA	
PSA	
Approach - EDDF-ILS 25R	
PSA	359° 0.0NM FL110

PAGE MENU

OPTIONS

- Activate Leg
- Show Chart
- Store Flight Plan
- Invert Flight Plan
- Delete Flight Plan
- Load Airway
- Collapse Airways**
- Remove Departure

Press the FMS CRSR knob to return to base page

ACTIVE FLIGHT PLAN LOWW / EDDF			
	DTK	DIS	ALT
WW178	270°	25.5NM	_____FT
SITNI	269°	31.1NM	_____FT
Airway - UL851.LNZ			
LNZ	287°	31.2NM	_____FT
Airway - T161.PSA (collapsed)			
PSA	359°	0.0NM	FL110
Approach - EDDF-ILS 25R			
PSA	359°	0.0NM	FL110
CHA	286°	12.5NM	_____FT
DF080	285°	7.2NM	_____FT
DF084	015°	5.0NM	_____FT



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FPL Composition with Airways



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FPL composition with AWYs LOWW SITNI L851 LNZ T161 PSA EDDF



Maybe that's your starting point:

Departure
and
Destination

Or when building the FPL sequentially:

STORED FLIGHT PLAN			
03	LOWW / EDDF		
		DTK	DIS
			ALT
	LOWW		
	EDDF	290°	336NM

STORED FLIGHT PLAN			
03	----- / LOWW		
		DTK	DIS
			ALT
	LOWW		



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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF



You continue
with loading a SID,
SITNI3A in this case

Departure - LOWW-RW11.SITN3A.SITNI				
RW11				_____FT
→ WW100	113°	16.6NM	1000FT	
WW101	189°	4.0NM	_____FT	
WW275	268°	15.0NM	4000FT	
WW178	269°	25.5NM	_____FT	
SITNI	268°	31.1NM	_____FT	
Enroute				
EDDF	295°	274NM	_____FT	

Departure - LOWW-RW11.SITN3A.SITNI				
RW11				
WW100	113°	3.0NM	1000FT	
WW101	189°	4.0NM		
WW275	268°	15.0NM	4000FT	
WW178	269°	25.5NM		
SITNI	268°	31.1NM		



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Flight Plan Composition



- Do you remember?
- A new waypoint is inserted above the highlighted waypoint

NORTH UP	
ACTIVE FLIGHT PLAN	
LOWG / LOWW	
WPT	DTK DIS
LOWG	
→ PUBEG	014° 34.5NM
LOWW	043° 51.4NM

This applies also to adding Airways!



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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF



Therefore we need one more point in the FPL sequence:

It can be EDDF,

Departure - LOWW-RW11.SITN3A.SITNI			
RW11			_____FT
→ WW100	113°	16.6NM	1000FT
WW101	189°	4.0NM	_____FT
WW275	268°	15.0NM	4000FT
WW178	269°	25.5NM	_____FT
SITNI	268°	31.1NM	_____FT
Enroute			
EDDF	295°	274NM	_____FT

or the empty point,

highlight it with the cursor

Departure - LOWW-RW11.SITN3A.SITNI			
RW11			
WW100	113°	3.0NM	1000FT
WW101	189°	4.0NM	
WW275	268°	15.0NM	4000FT
WW178	269°	25.5NM	
SITNI	268°	31.1NM	

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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF



Now you turn the small FMS knob clockwise one click



Which offers you the „LD AIRWY“ softkey



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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF



Or you call the **MENU** and use the option **Load Airway**
(which is the only way to do it on the PFD)



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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF



Now you select the Airway



And then you select the exit point



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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF

The airway is now entered into the Flight Plan

You highlight the next waypoint

or „empty“ point
and continue with the procedure as before

Departure - LOWW-RW11.SITN3A.SITNI				
RW11				_____FT
→ WW100	113°	2.8NM	1000FT	
WW101	189°	4.0NM	_____FT	
WW275	268°	15.0NM	4000FT	
WW178	270°	25.5NM	_____FT	
SITNI	268°	31.1NM	_____FT	
Airway - UL851.LNZ				
LNZ	287°	31.2NM	_____FT	
EDDF	296°	243NM	_____FT	

WW101	189°	4.0NM		
WW275	268°	15.0NM	4000FT	
WW178	269°	25.5NM		
SITNI	268°	31.1NM		
Airway - L851.LNZ				
LNZ	287°	31.2NM		

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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF

You select the Airway

And then you select the exit point

ENTRY	
LNZ	
AIRWAY	
T161	
L604	
M141	
233	
T161	
UL604	
UM141	
UT161	
290° 33.1NM	
LOAD?	

ENTRY	
LNZ	
AIRWAY	
T161	
EXIT	
PSA	
LNZ	
SUBEN	
ABABI	
AKINI	
NIMDI	
ERNAS	
REDNI	
ASPAT	
ROTAX	
TALAK	
LEPSA	
PSA	
TALAK	
LEPSA	
PSA	
290° 33.1NM	
290° 16.7NM	
289° 35.1NM	
278° 19.7NM	
277° 16.6NM	
316° 19.2NM	
314° 20.7NM	
313° 33.4NM	
298° 14.0NM	
283° 9.7NM	
LOAD?	

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FPL composition with AWYs

LOWW SITNI L851 LNZ T161 PSA EDDF

Again the airway is entered
into the Flight Plan

From here you can
continue with loading an
Approach into Frankfurt

ACTIVE FLIGHT PLAN			
LOWW / EDDF			
	DTK	DIS	ALT
Airway - T161.PSA			
SUBEN	290°	33.1NM	_____FT
ABABI	290°	16.7NM	_____FT
AKINI	289°	35.1NM	_____FT
NIMDI	278°	19.7NM	_____FT
ERNAS	277°	16.6NM	_____FT
REDNI	316°	19.2NM	_____FT
ROTAX	314°	20.7NM	_____FT
TALAK	313°	33.4NM	_____FT
LEPSA	298°	14.0NM	_____FT
PSA	283°	9.7NM	_____FT



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FPL composition with AWYs

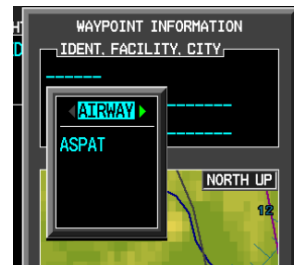
LOWW SITNI L851 LNZ T161 PSA EDDF

■ Inserting „Optional Waypoints“

AIRWAY	
T161	
EXIT	
PSA	
LNZ	
SUBEN	290° 33.1NM
ABABI	289° 16.7NM
AKINI	289° 35.1NM
NIMDI	277° 19.7NM
ERNAS	277° 16.6NM
REDNI	316° 19.2NM
ASPAT	314° 20.7NM
ROTAX	
TALAK	
LEPSA	
PSA	

LOAD

SUBEN	SUBEN
ABABI	ABABI
AKINI	AKINI
NIMDI	NIMDI
ERNAS	ERNAS
REDNI	REDNI
ROTAX	ROTAX
TALAK	TALAK
LEPSA	LEPSA
PSA	PSA

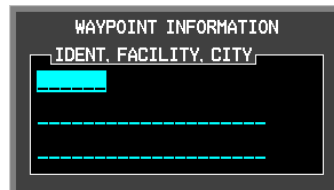


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WPT entry logic



Small FMS knob clockwise:
Enter WPT
Small FMS knob counter-clockwise:
WPT Submenu:



**A special case when loading
an airway into the FPL:**



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Approach Procedures in the Flight Plan



Approach Procedures

■ Approach

- „loading“ and „activating“ has different meaning !
- Loading:
 - adds procedure to the flight plan, ready for quick activation
- Activating:
 - overrides en-route part of flight plan
 - Proceeds direct to approach part
 - Initiates automatic CDI scaling



Approach Procedures



■ Approach

- Select → Load → Activate
- or
- Select → Activate



Approach Procedures



Loading an ILS Approach:

If you navigate on GPS:

- ILS frequency is put into the „Active“-field!

If you navigate on VHF-NAV:

- ILS frequency is put into the „Standby“-field
- You have to flip-flop it manually to „active“



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FPL „along track offset“



ACTIVE FLIGHT PLAN:			
KMHC / KDOS			
	DTK	DIS	ALT
TIFTO	---	---	---
Enroute			
TGP	---	---	---
Airway - V4.SLN			
SLN -10NM	258°	21.9NM	12000FT
SLN	259°	10.0NM	12000FT
Airway - V244.LAA			
HVS	262°	77.7NM	12000FT
LAA	251°	165NM	12000FT
Airway - V263.TBE			
TBE	210°	71.1NM	12000FT

CURRENT VNV PROFILE:			
ACTIVE VNV WPT 9000FT at HABUK 10F			
VS TGT	-467FPM	FPA	-1.5°
VS REQ	---	TIME TO TOD	02:25
V DEV	---		

Offset WPT

Can be used to reach a specified altitude before or after the specified flight plan waypoint

Offset WPT may be in front or after selected WPT

Distance up to 99 NM

On PFD: call the menu

On MFD: ATK OFST softkey



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FPL „parallel track“



- Available on the current flight plan, offset 1 to 50 NM left or right
- Parallel track not available:
 - for „direct to“
 - first leg of departure route
 - approach leg
 - unreasonable route geometry
- Parallel track cancelled automatically:
 - using „direct to“
 - activating an approach
 - editing the flight plan
- VNV not available with parallel track



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FPL „parallel track“

Call the menu

Select Parallel Track

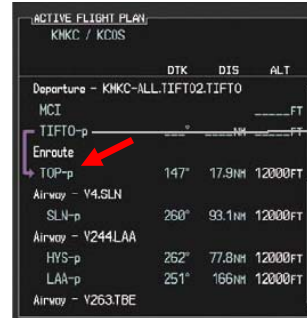


Enter parameters

Activate Parallel Track



Parallel track is indicated by a -p



To cancel „parallel track“:

Call the menu, call the parallel track window, activate CANCEL PARALLEL TRACK



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Vertical Navigation (VNV)



Top of Descent message:

Top of Descent Message

NAV1	113.00 ↔ 116.70	PUB	→ KAFF	TOD within 1 minute	118.900	133.000	COM1
NAV2	116.70 ↔ 108.80	FCS			135.325 ↔ 124.300		COM2

Comes together with aural alert



Vertical Navigation (VNV)

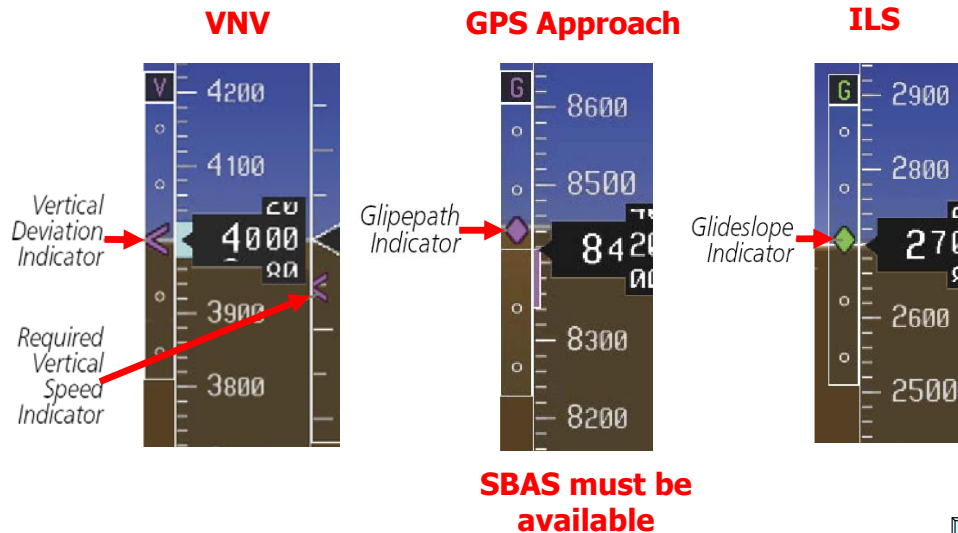


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Vertical Navigation (VNV)

- Observe the different meaning:
 - „GP” mode – Glide Path – for vertical approach guidance (available with WAAS only)
 - „GS” mode – Glide Slope – for ILS navigation

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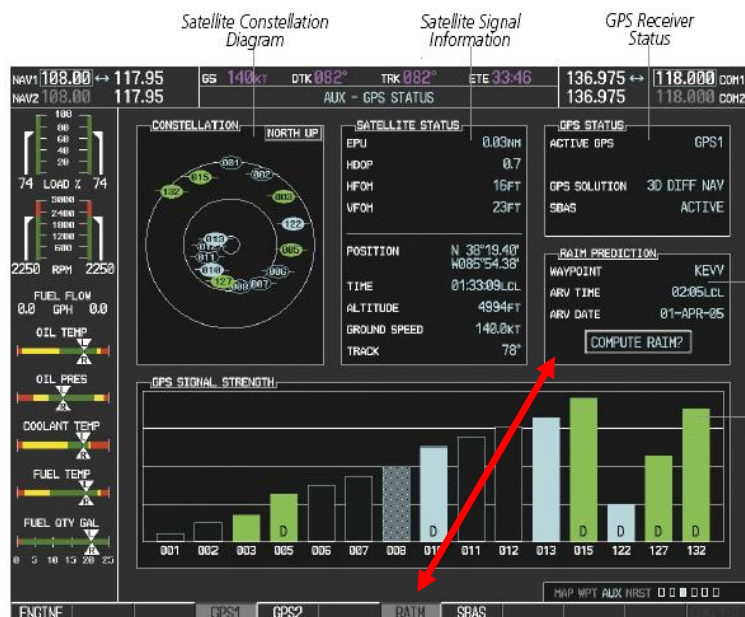
Vertical Navigation (VNV)

Vertical Navigation (VNV) provides descent guidance only, based on barometric altitude.

For vertically guided **approaches** (LPV, LNAV/VNAV, LNAV+V) **SBAS** is required!



GPS status page (AUX 3)



RAIM
Availability
Prediction

Satellite
Signal
Strength
Bars



GPS status page (AUX 3)



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Vertical Navigation (VNV)

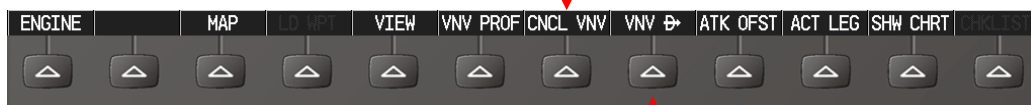
- With VNV new vertical AFCS modes:
 - **VPTH** Vertical Path Tracking
 - **ALTS** Selected Altitude Capture
 - **ALTV** VNV Target Altitude Capture

Vertical Navigation (VNV)



↑
Enable VNV

↓
Cancel VNV



↑
VERTICAL DIRECT TO

Like a horizontal DIRECT TO bypasses all intermediate waypoints, the VERTICAL DIRECT TO bypasses intermediate altitude constraints and makes a straight descent to the altitude constraint at the waypoint selected for the VERTICAL DIRECT TO

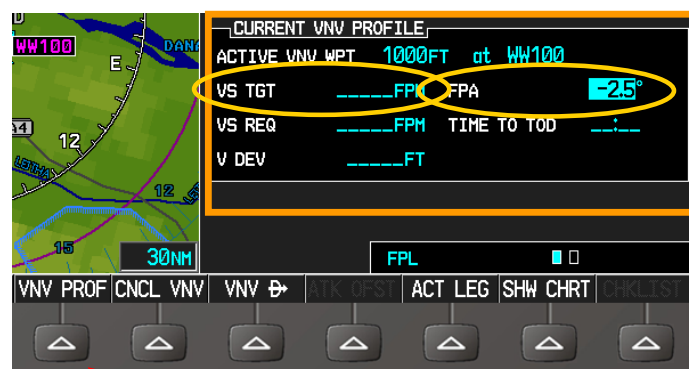


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Vertical Navigation (VNV)



↑
to modify VNV profile



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Vertical Navigation (VNV)

Altitude constraints in a Flight Plan

ACTIVE FLIGHT PLAN			
KIXD / KDFW			
	DTK	DIS	ALT
KARLA	221°	11.7NM	13000FT
COVIE	221°	9.0NM	12400FT
LEMYN	220°	8.0NM	9900FT
Approach - KDFW-RNAV 17L GPS LPV			
RIVET iaf	259°	18.8NM	4000FT
DRAAK	176°	3.3NM	2000FT
INWOD	176°	3.2NM	3000FT
MENOL faf	176°	3.9NM	2300FT
RW17L mop	176°	5.3NM	
9900FT	174°	0.8NM	990FT
POLKE			



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Vertical Navigation (VNV)

- Altitude constraints may be
 - „Designated“ ← Used in giving vertical speed and deviation guidance
 - or
 - „Reference only“
- Altitude constraints may be
 - Manually entered
 - or
 - Retrieved from database



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Vertical Navigation (VNV)



- Entering altitude constraints at Flight Plan waypoints:
 - ALTITUDE: small FMS knob to the right
 - FLIGHT LEVEL: small FMS knob to the left
 - If WPT is an airport: after ENT you can choose „MSL“ or „AGL“



Vertical Navigation (VNV)



- „Designating“ a constraint:
 - ENT key
- Cancelling or modifying a constraint:
 - CLR key



Vertical Navigation (VNV)

- Colour coding of altitude constraints:
 - White:
 - Reference only
 - Blue:
 - Designated
 - Blue subdued:
 - Unuseable because of invalid constraint condition
- Font-size coding of altitude constraints:
 - Large:
 - Manually entered or calculated by the system
 - Small:
 - Database retrieved (or manually entered and matching database)



Vertical Navigation (VNV)

Altitude constraint coding

	White	Blue
LARGE	Manual or calculated reference	Manually designated
SMALL	Database reference	Database designated



Vertical Navigation (VNV)



Altitude constraint examples

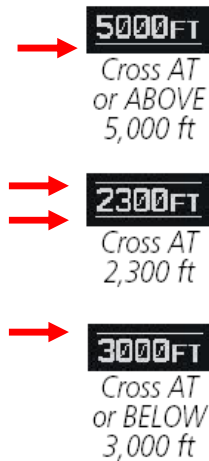


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Vertical Navigation (VNV)



- On some „leg types“ Vertical Navigation is not available:
 - CA Course to altitude
 - CI Course to intercept
 - FA Fix to altitude
 - FM Fix to manual termination
 - HA Hold to altitude
 - HM Hold to manual termination
 - PI Procedure turn to intercept
 - VA Heading to altitude
 - VD Heading to DME distance
 - VI Heading to next leg intercept
 - VR Heading to radial termination
 - VM Heading to manual termination



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Never forget ...



FMS – ACFT „connection“

„Connection“ to ACFT

